



Klinikum rechts der Isar
Technische Universität München



MICCAI 2024

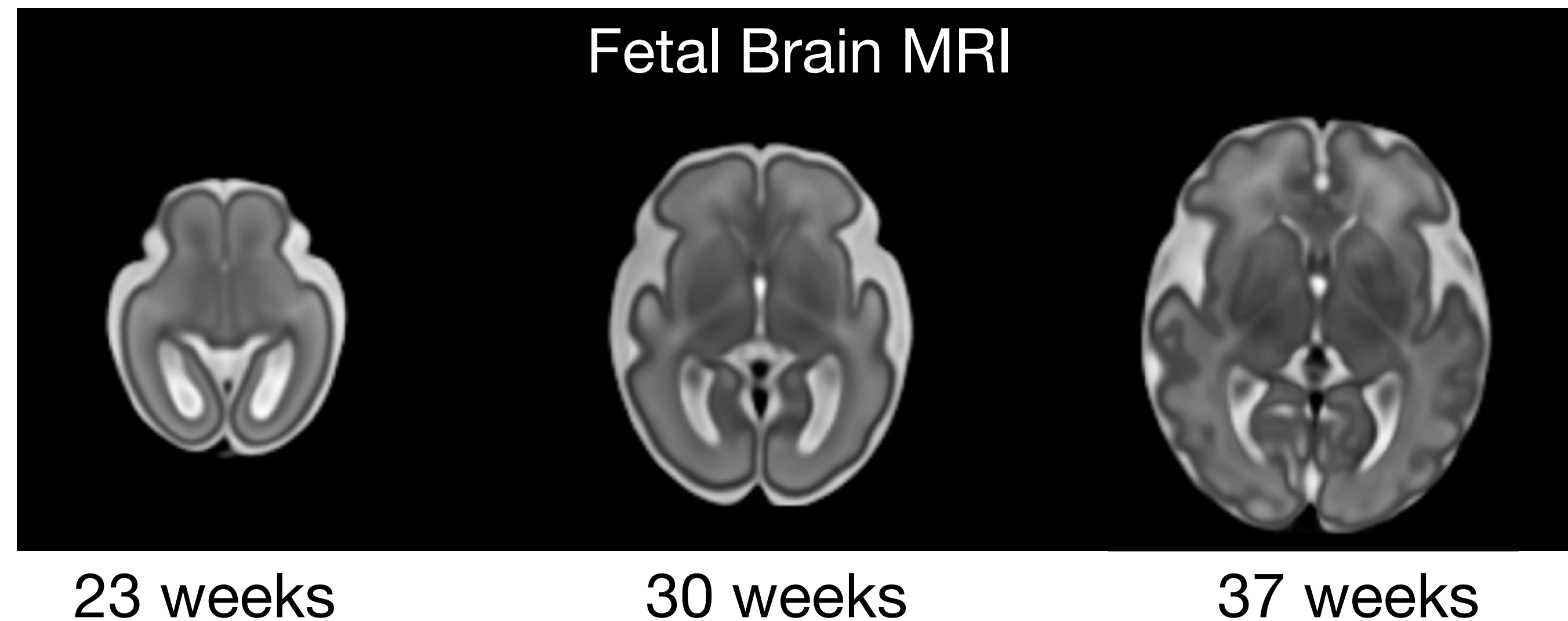
INR Tutorial

Maik Dannecker, 06.10.2024



Fetal Brain Development

Fetal brains undergo **rapid changes** throughout pregnancy involving different stages of growth and cortical folding. Understanding the spatio-temporal trajectory of fetal brain development is crucial for early identification of atypical brain patterns and potential neurodevelopmental disorders.



—> There is a need for spatio-temporal representations of the fetal brain.

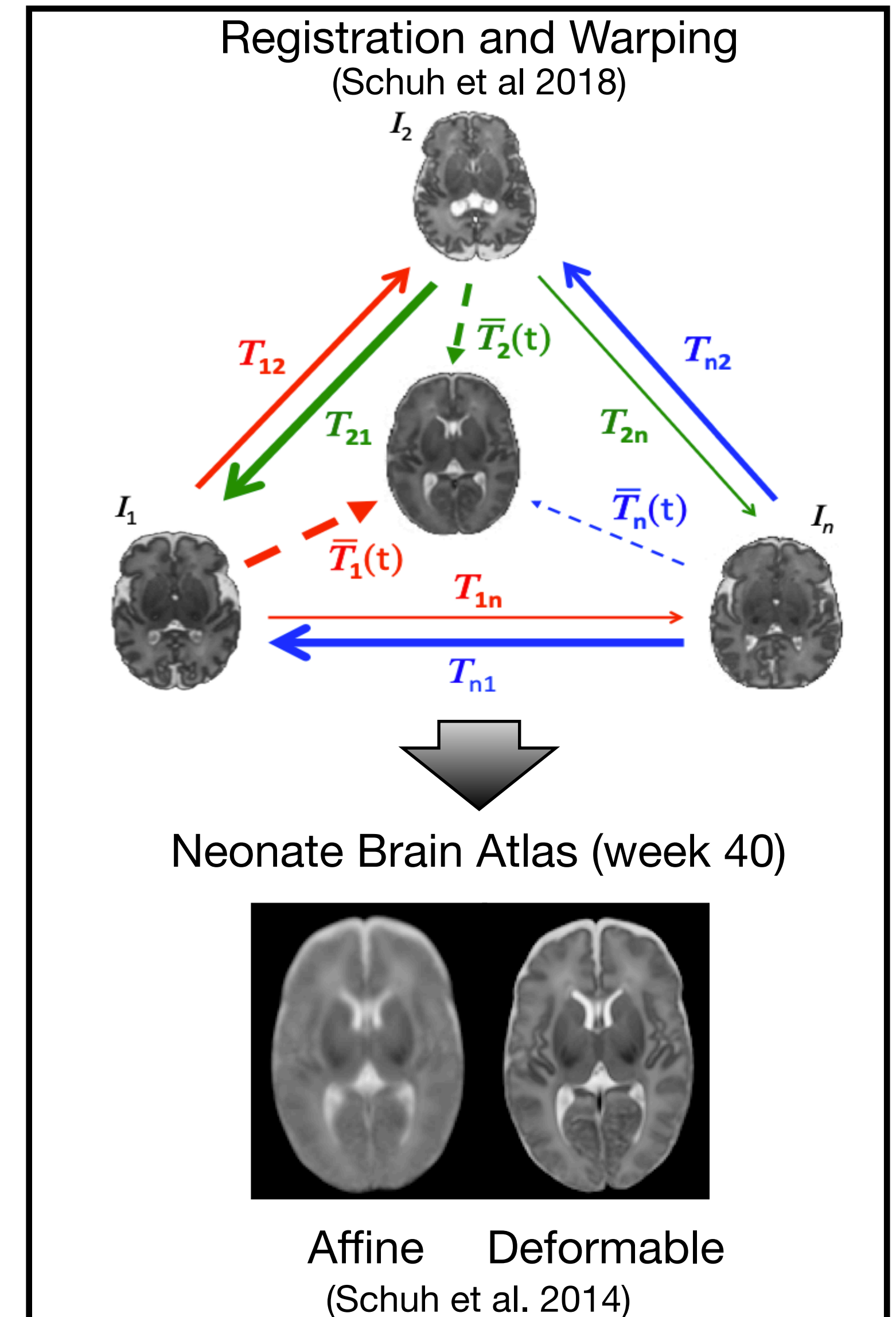
Atlases in Medical Imaging

- In medical imaging, an atlas serves as a ***normative representation*** of an anatomy (e.g. brain)
- Use-cases?
 - Reference for spatial alignment (registration)
 - Prior knowledge (e.g., for segmentation of brain tissue)
 - Group analysis (e.g., **identification of atypical brain patterns**)

(Classic) Atlas Construction

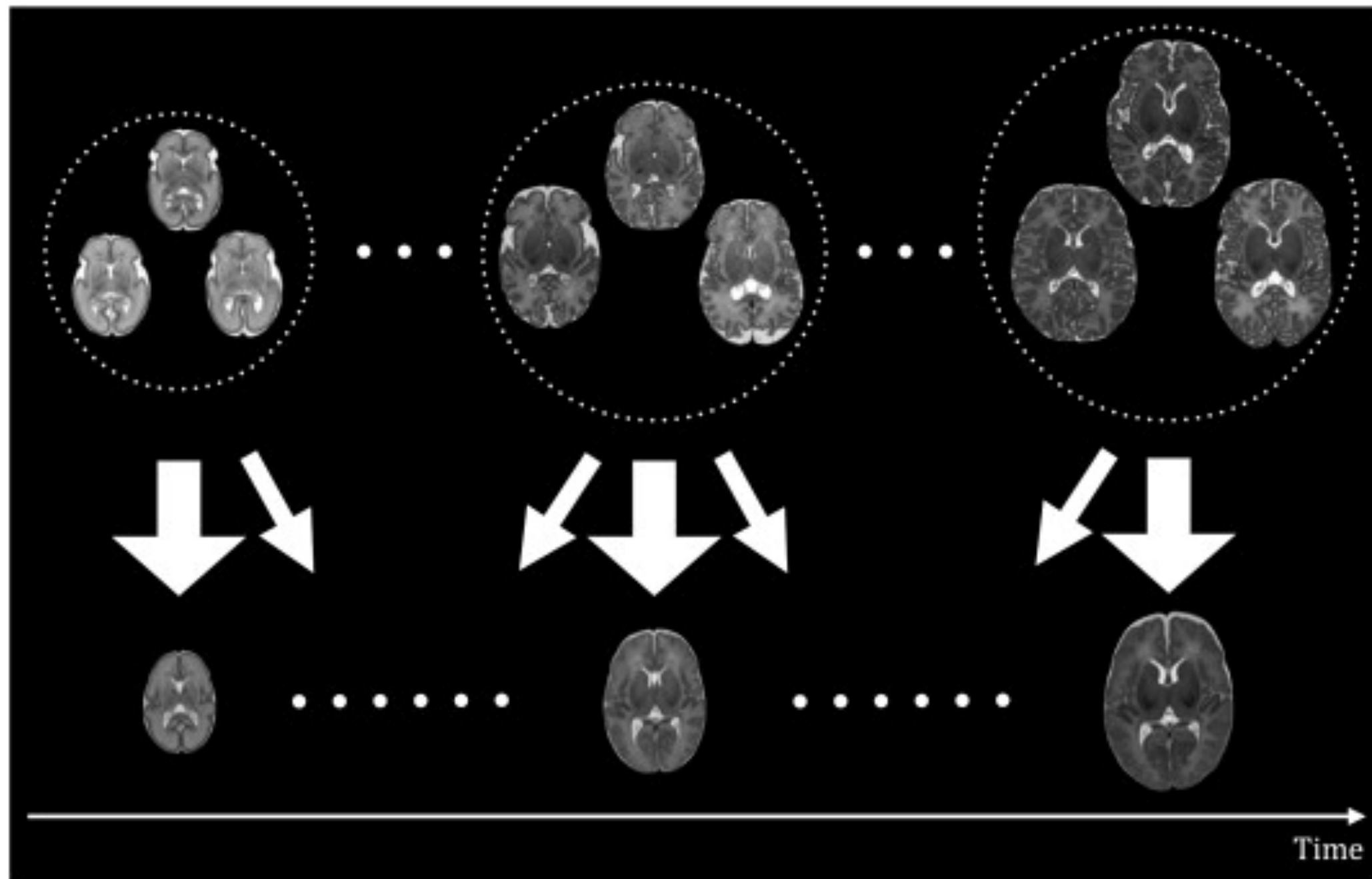
Atlas construction from a (normative) population:

- Trivial approach:
 - Mean over (aligned) images
 - > blurry due to high variability of images
- Registration approach:
 - Align images via affine or deformable registration
 - Apply *mean transformation* to warp images to common shape
 - > sharper delineation (variability in transformation not in the image)

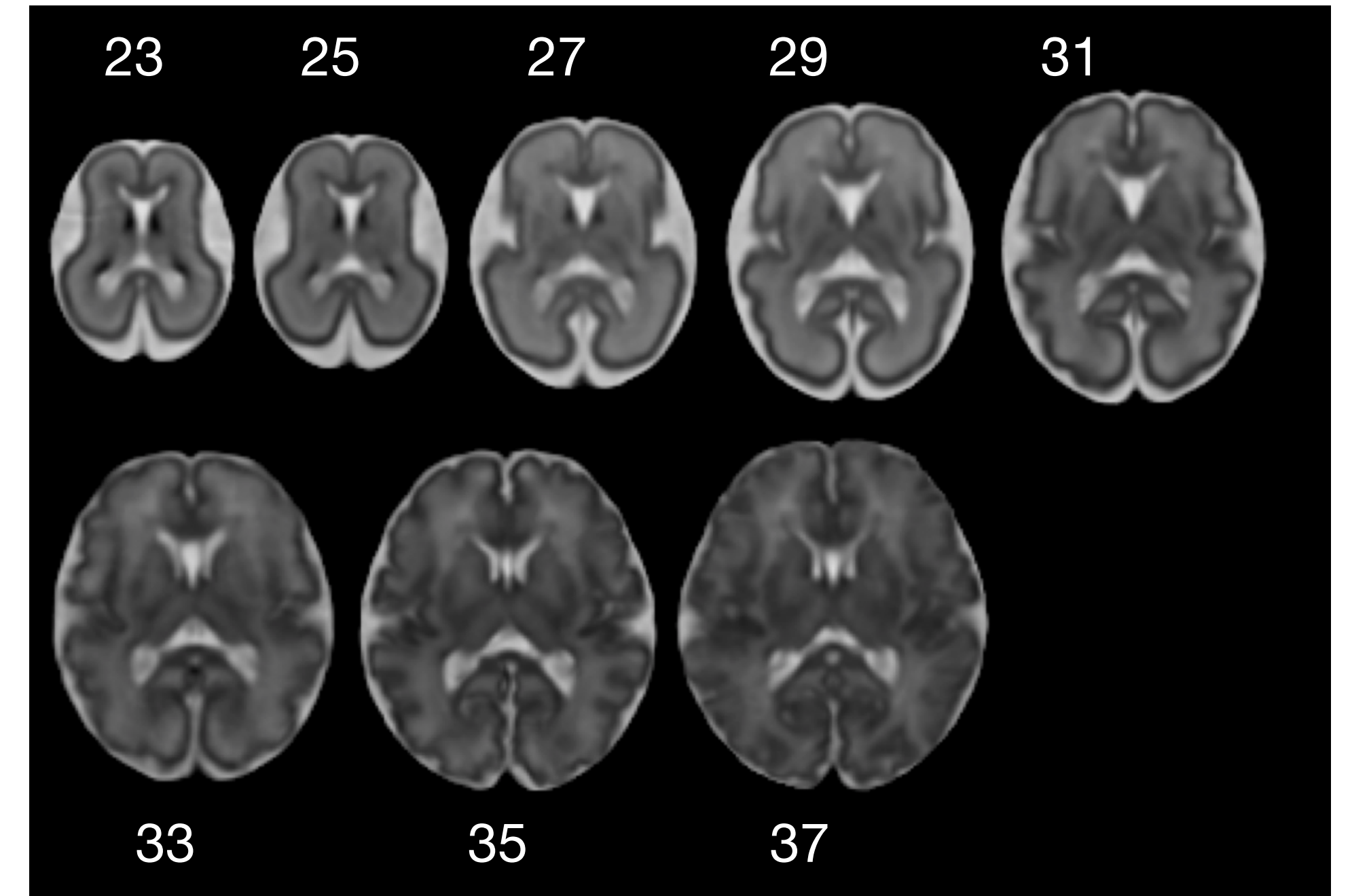


Fetal Brain Atlas

- Fetal brain MRI ranges from ~ 20 weeks gestational age (GA) to 38 weeks GA
 - > rapid brain development requires a dynamic atlas
 - > dynamic atlas requires sufficient data over time



(Serag et al. 2012)

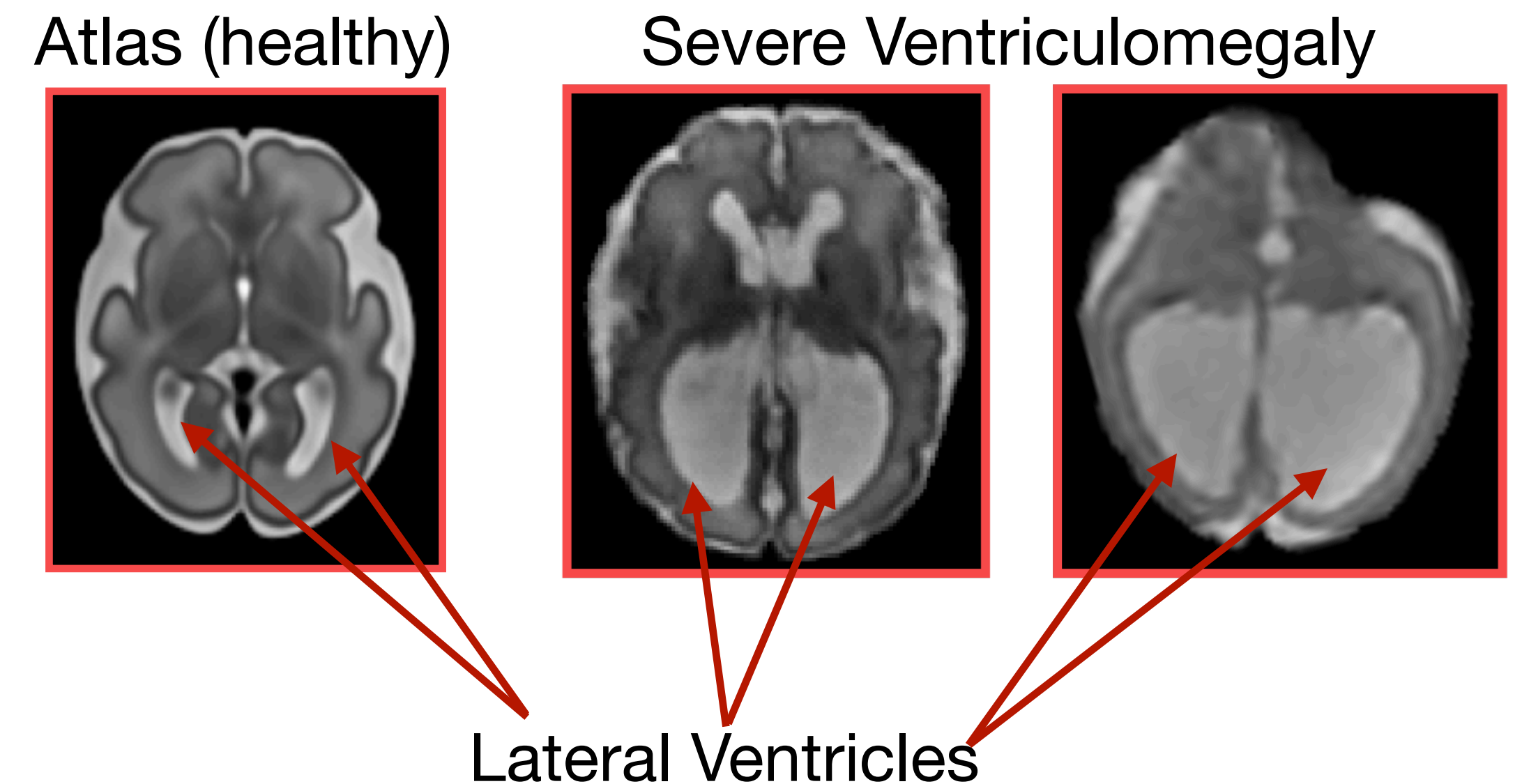


(brain-development.org (modified))

- Here, 80 subjects across age range used for atlas generation

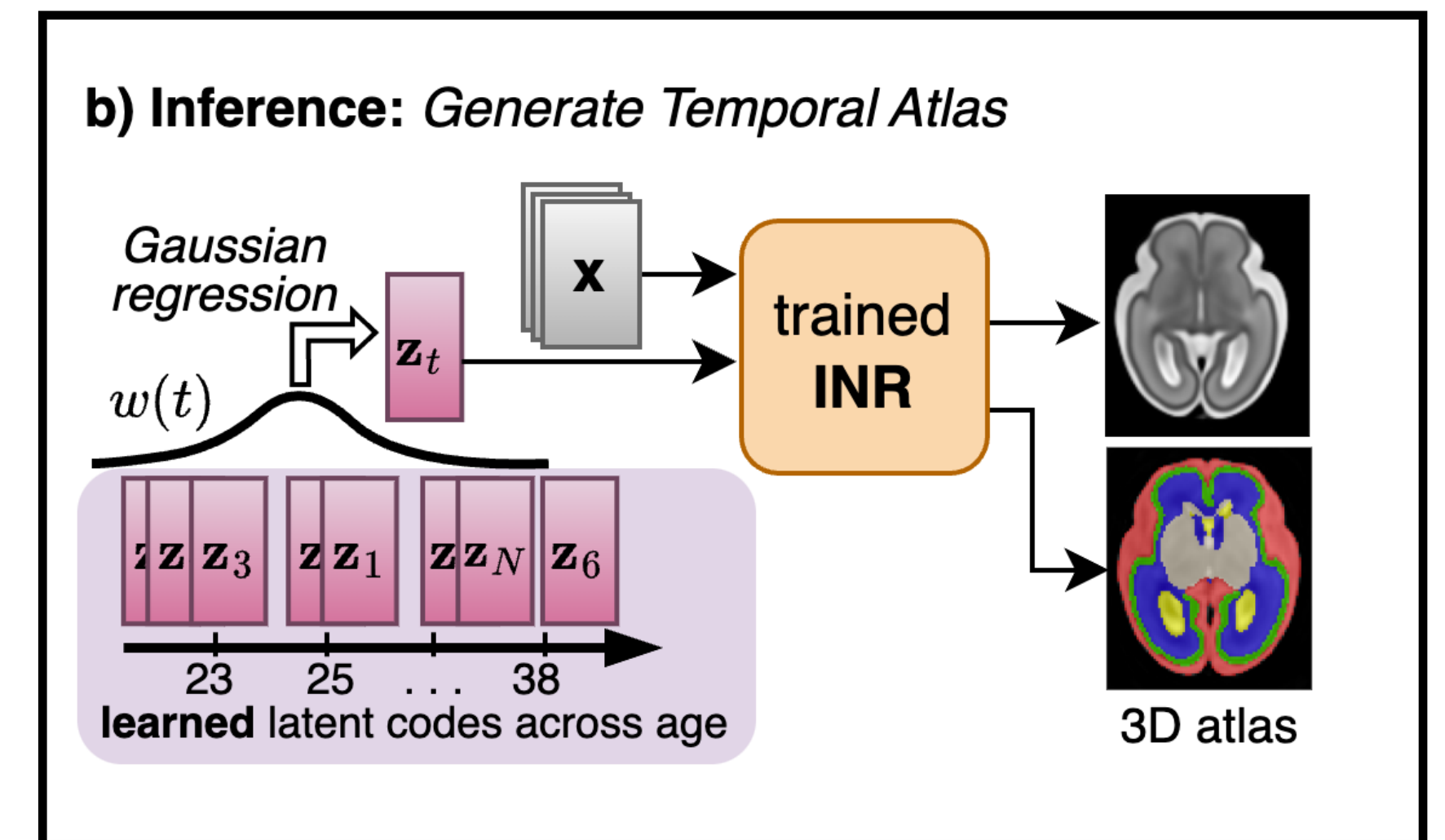
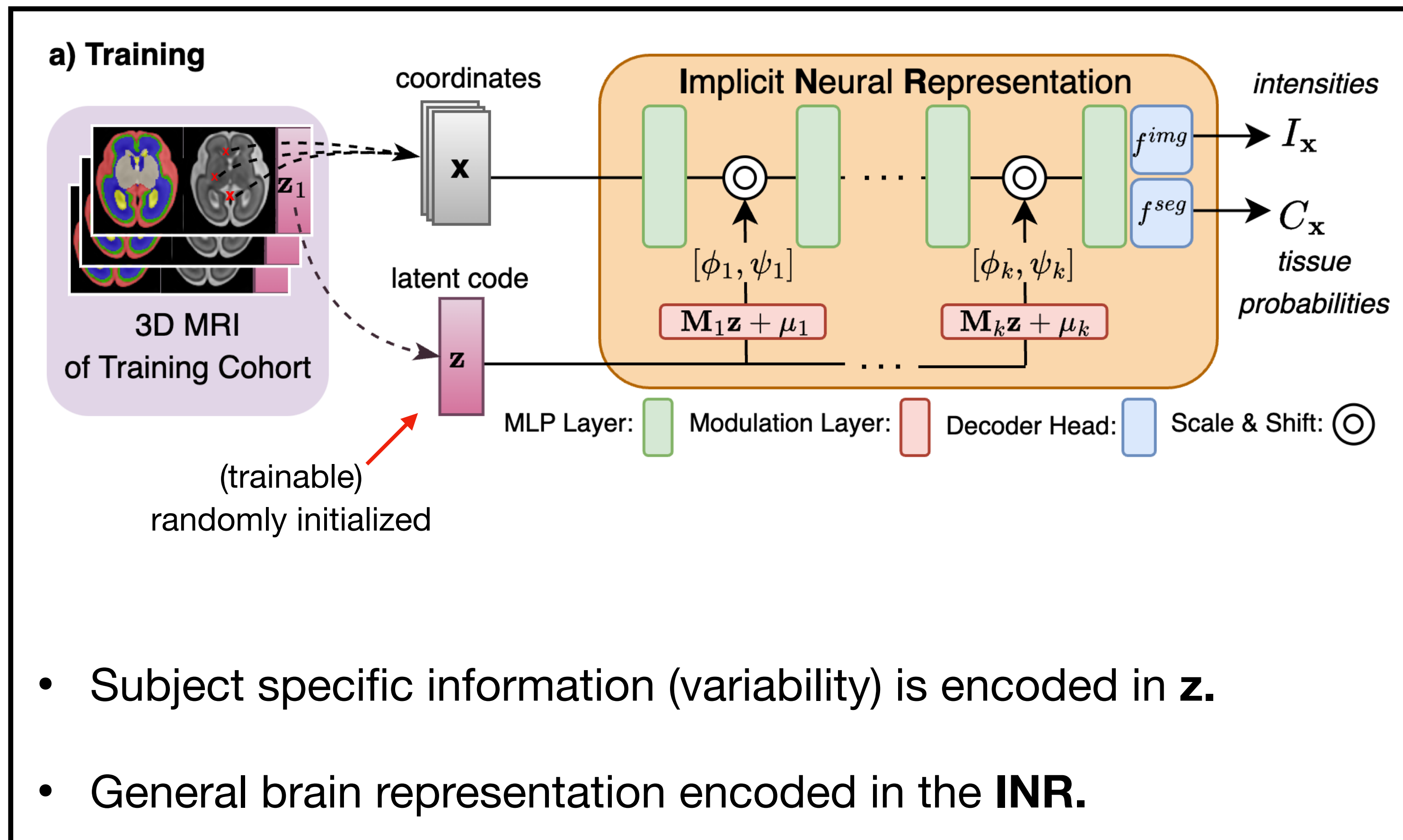
Fetal Brain Atlas for Pathological Data

- Most atlases focus on healthy data and do not adequately model pathologies such as Ventriculomegaly (VM)
- Classic methods are static models.
—> A new atlas is required for each use case.
- Public pathological data is rare, impeding atlas construction for pathology modeling. E.g., FeTA Dataset ~30 usable pathological scans, ~15 with VM.

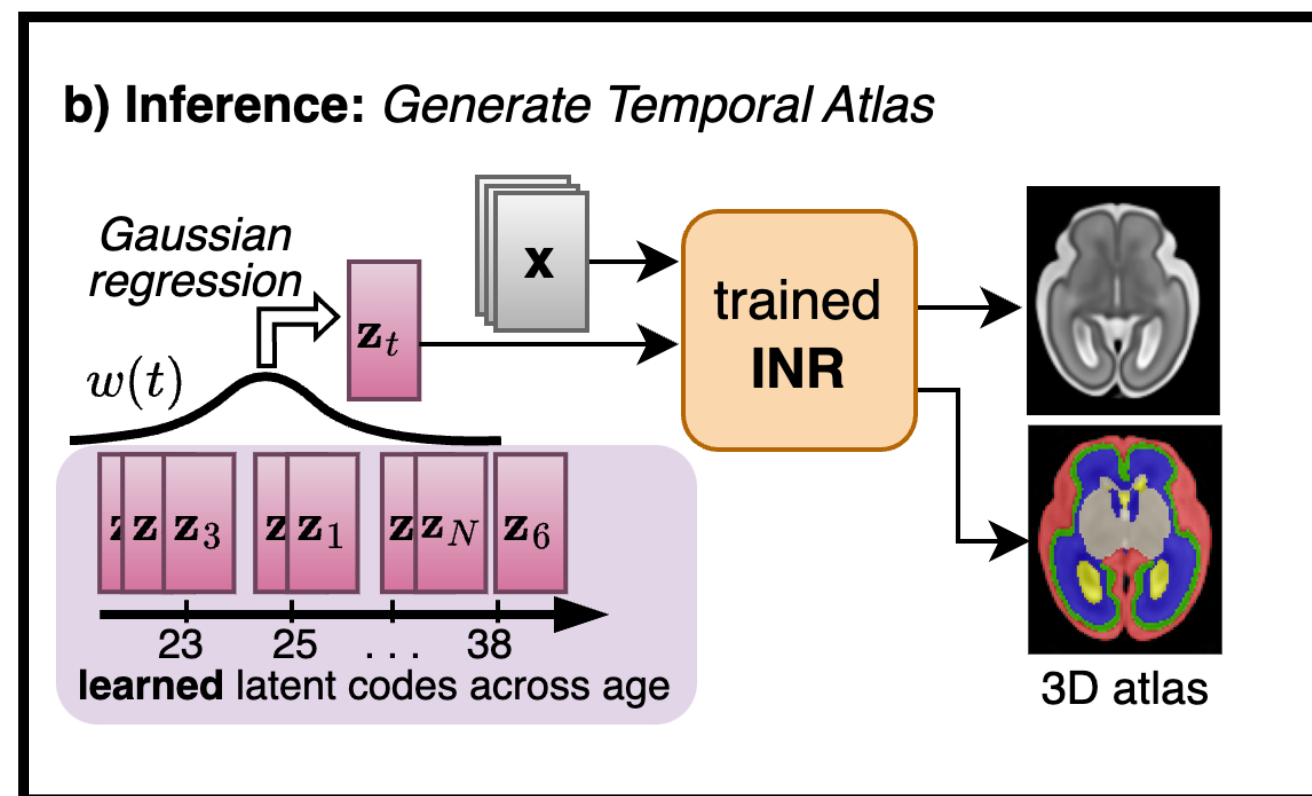


CINA: Conditional Implicit Neural Atlas

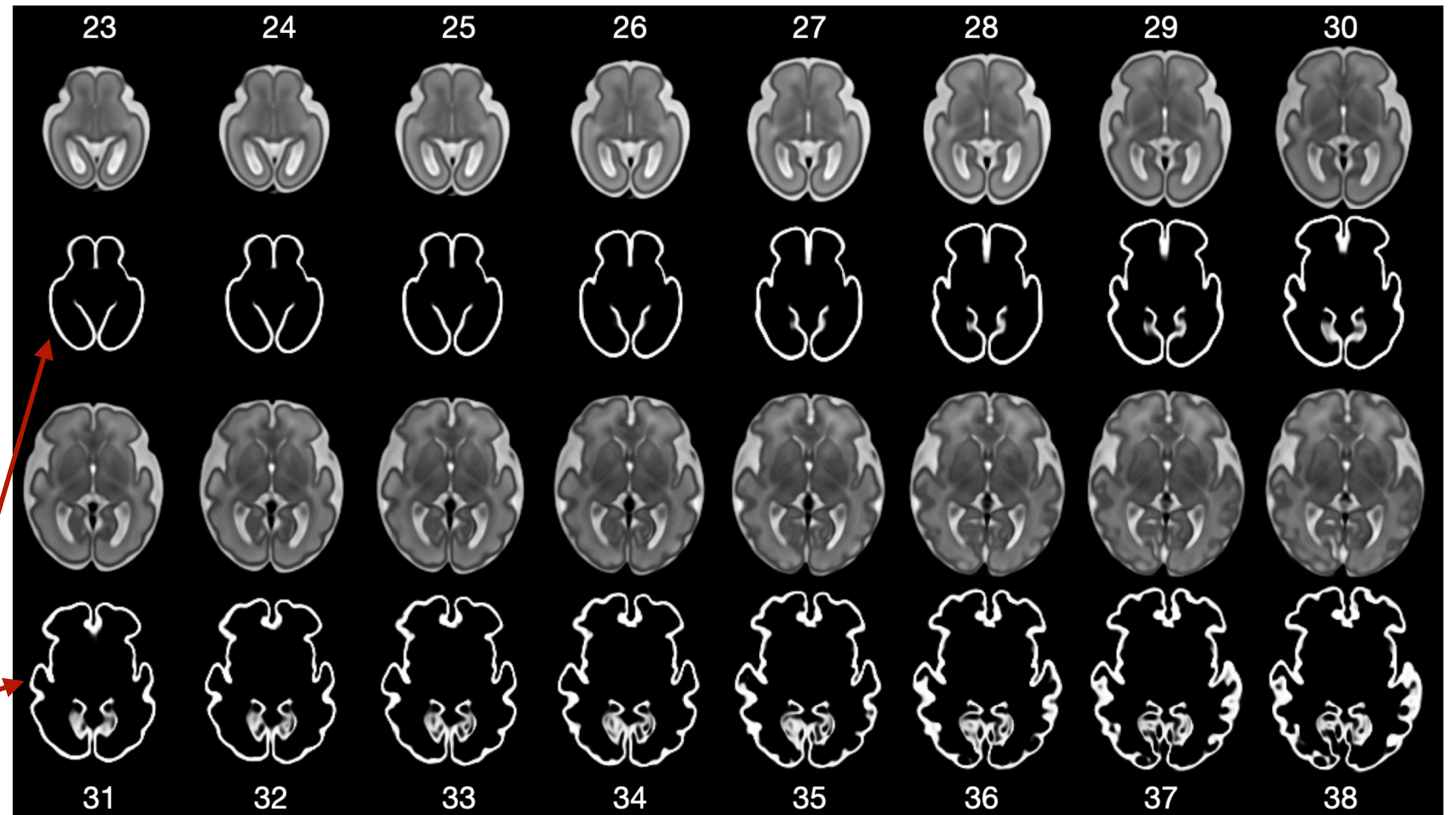
- Spatio-temporal atlas construction using conditional implicit neural representation.



Spatio-Temporal Atlas Construction with CINA



tissue probability map
(cortical grey matter)

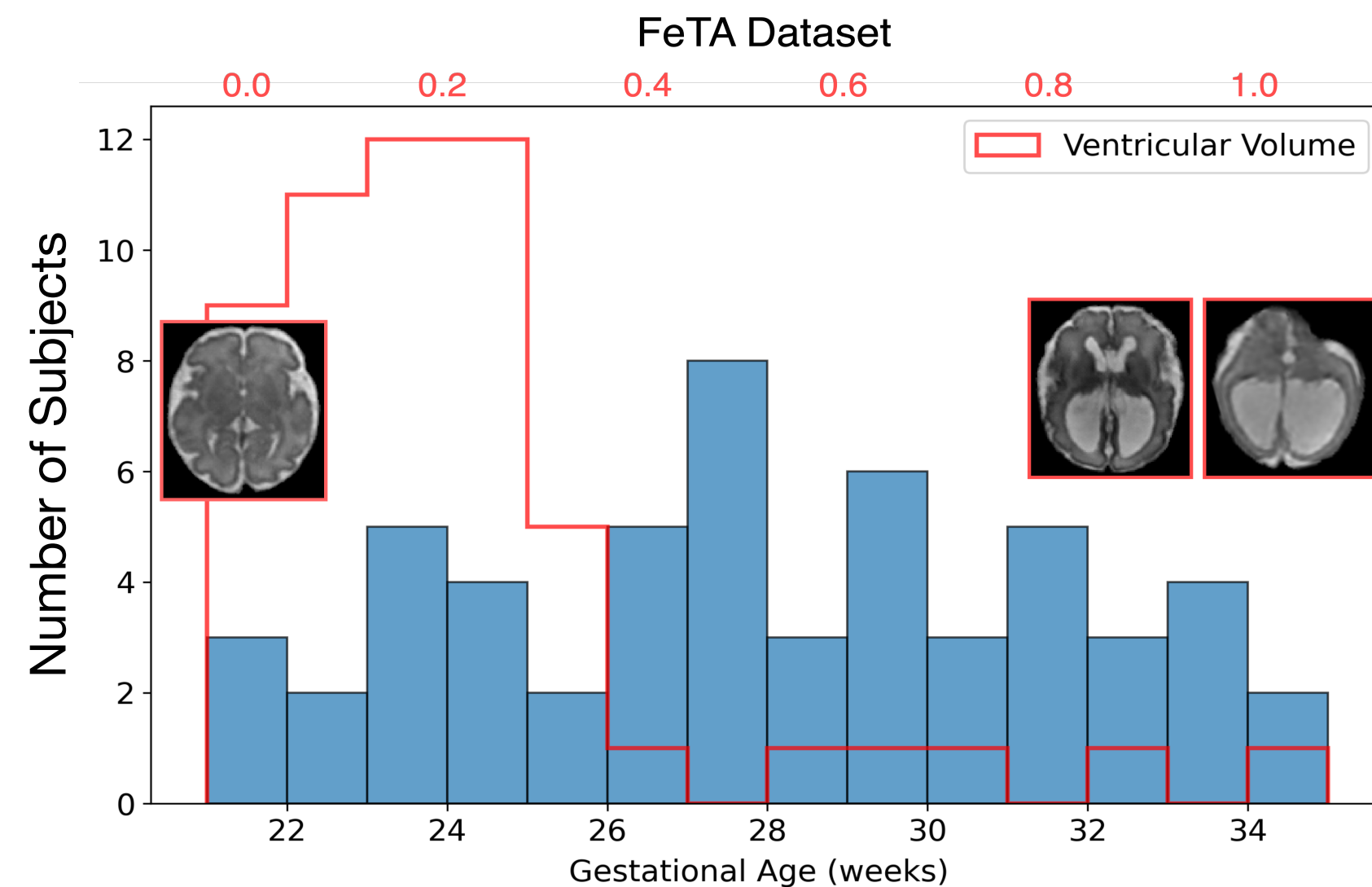


Fetal brain atlas of continuous spatial and temporal resolution.

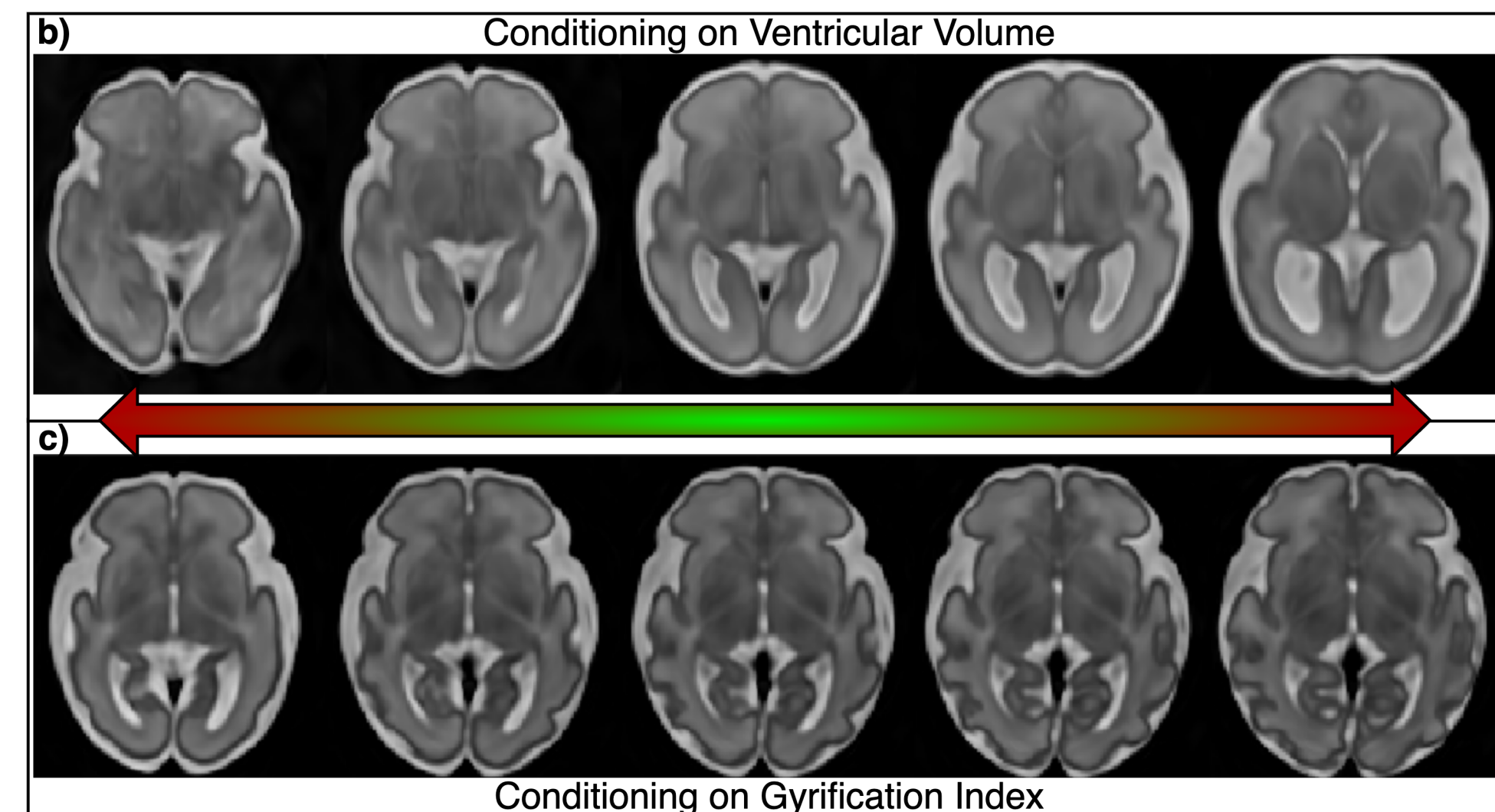
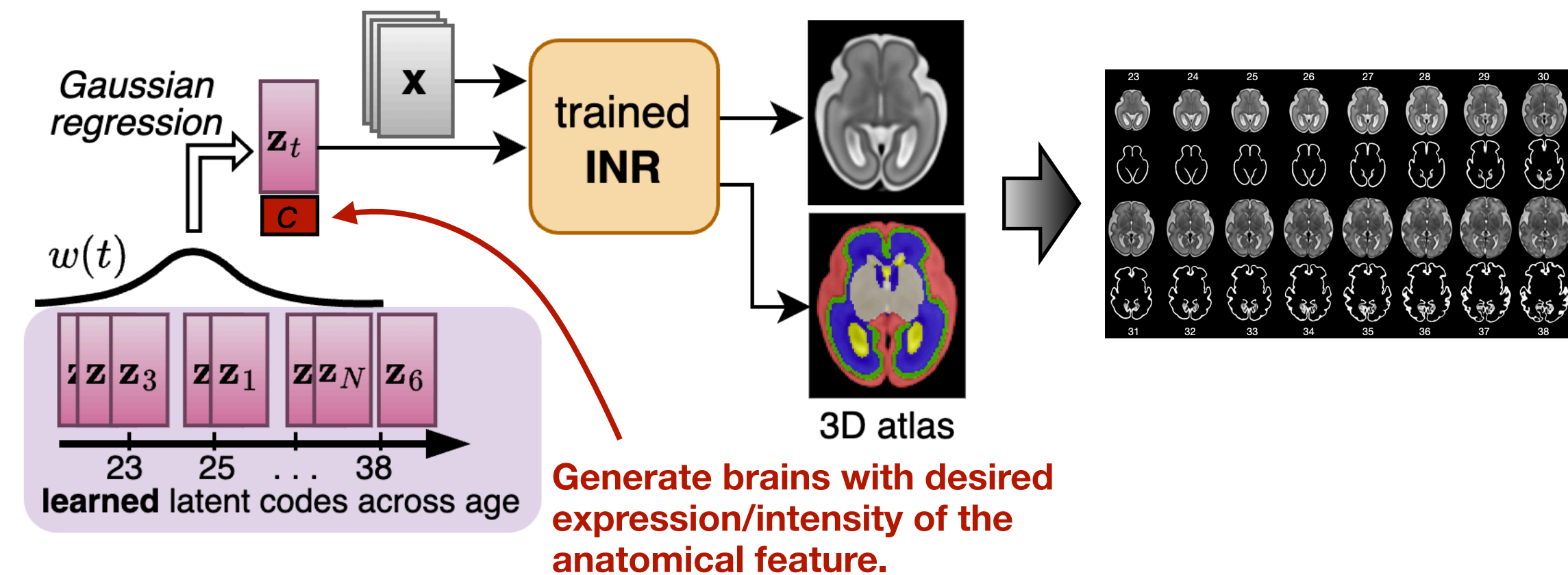
Conditioning on Anatomical Features

Explicitly condition on:

- Ventricular Volume
 - Add subject's ventricular volume as additional dimension to $\mathbf{z}$ during training.
- Gyrification
 - Add subject's gyrification index (degree of cortical folding) to $\mathbf{z}$.



b) Inference: Generate Temporal Atlas



Align Atlas to New Subjects

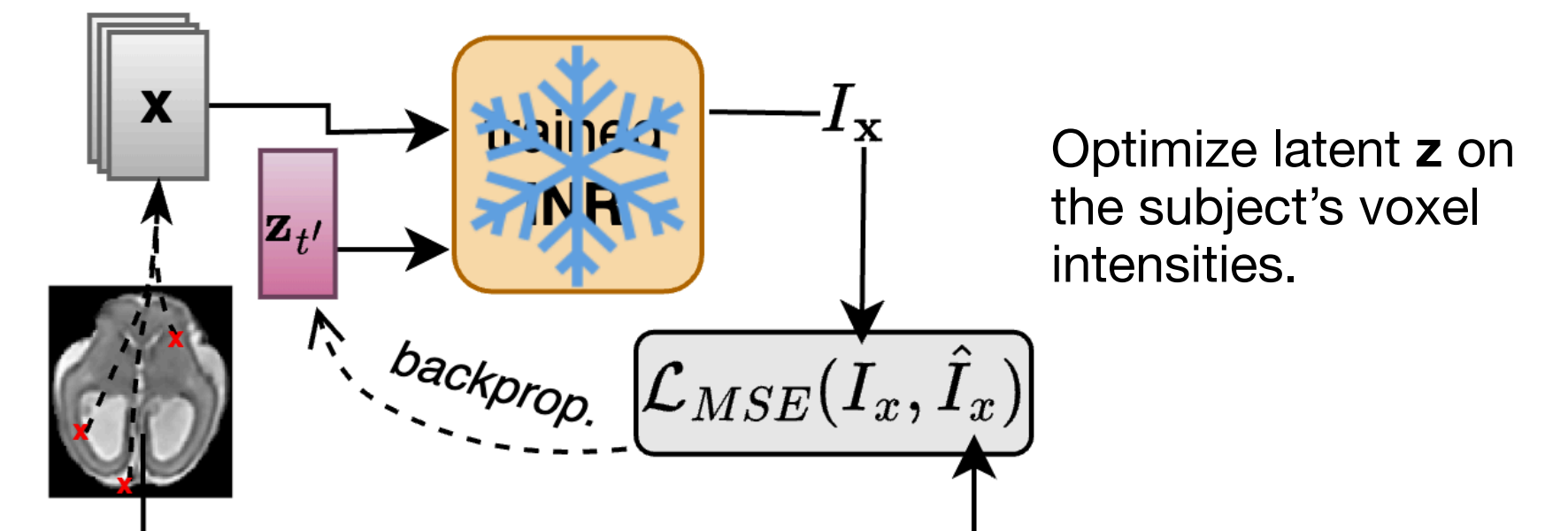
CINA:

- ▶ initialize new latent code $\mathbf{z}$.
- ▶ optimize $\mathbf{z}$ during test-time, freezing the INR and minimizing MSE-Loss on voxel intensities of new subject.
- ▶ predict brain tissue via single forward pass

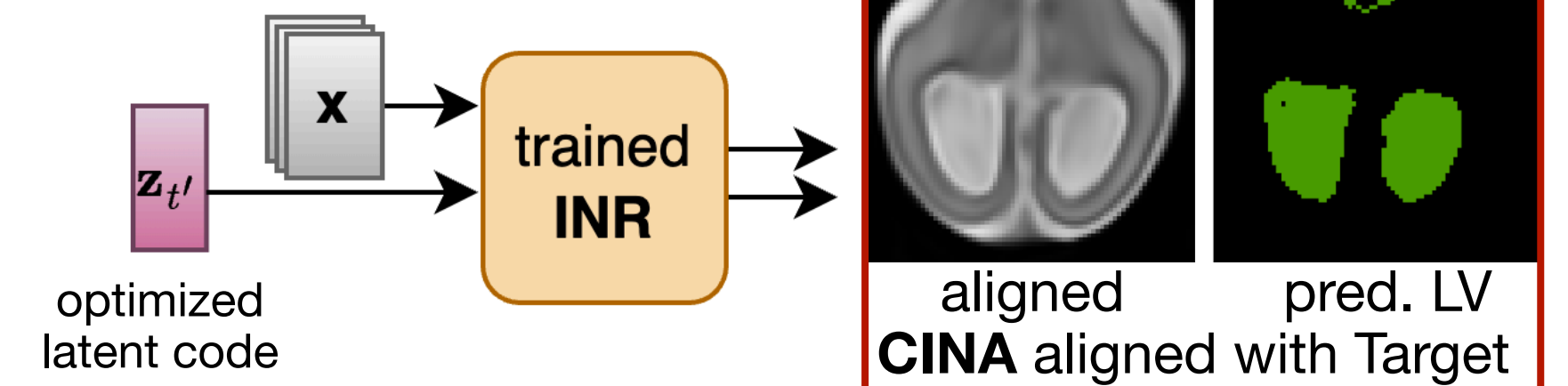
Classic methods:

- ▶ register atlas to new subject by finding optimal transformation T .
- ▶ project atlas tissue maps to new subject using T .

1) CINA: Test-Time Optimization on Target Subject



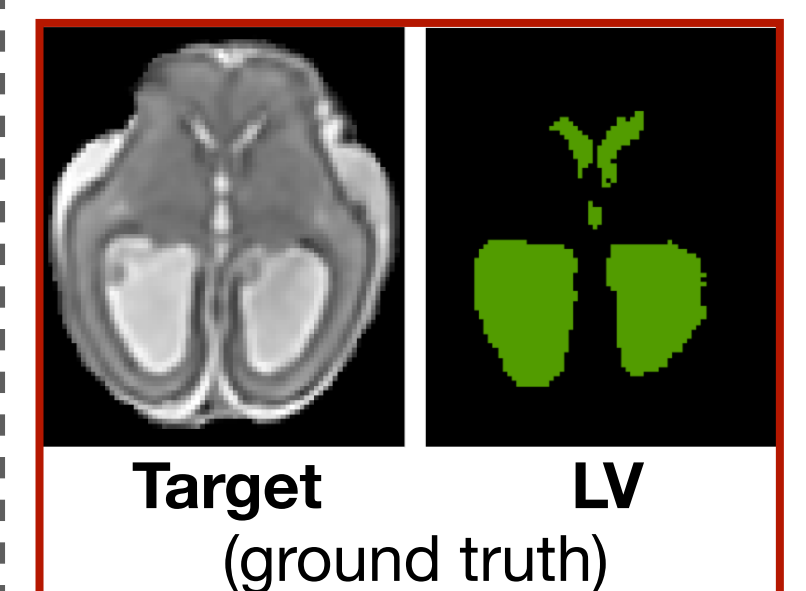
2) CINA: Prediction



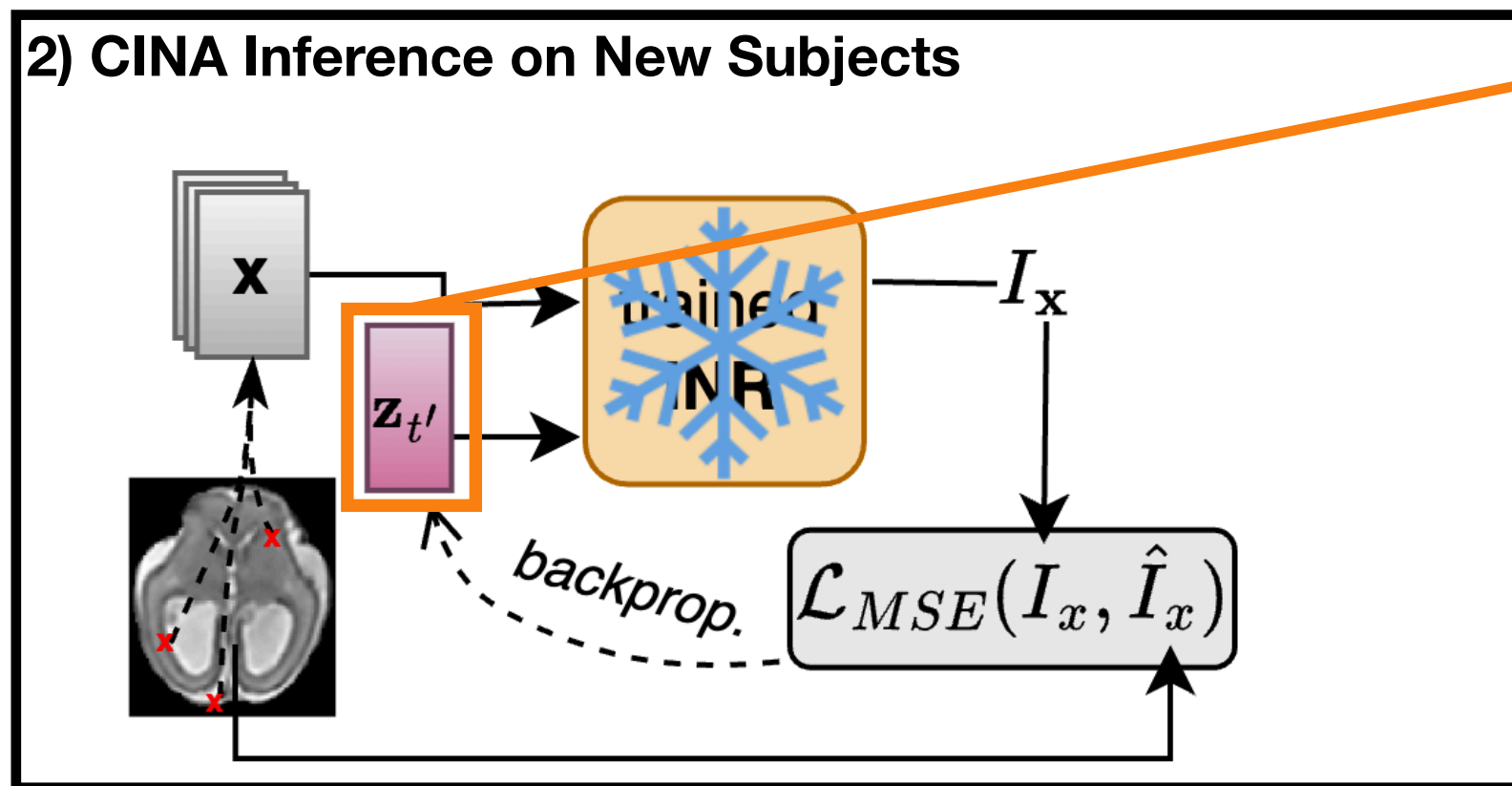
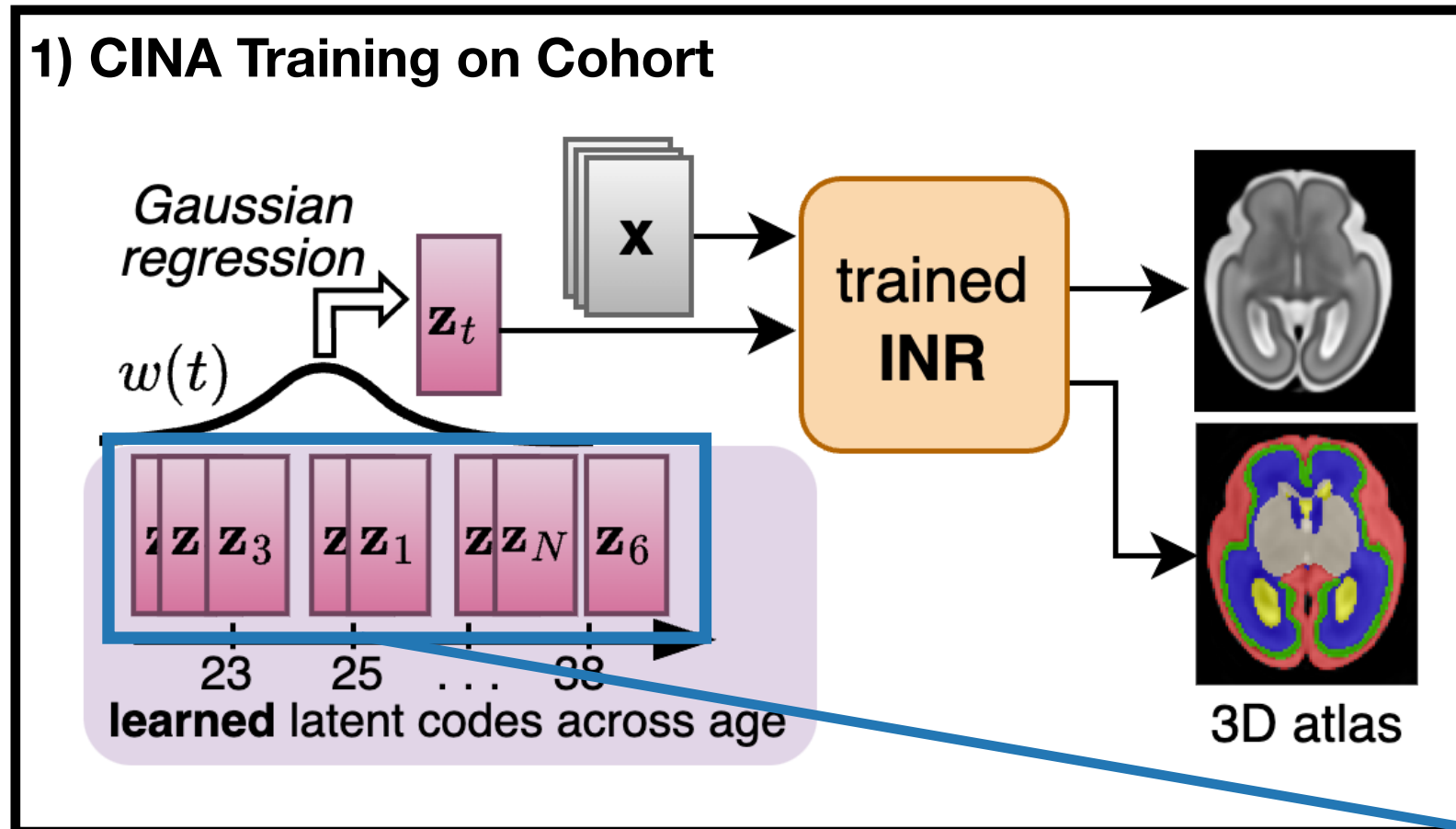
3) Classic Atlas: Prediction



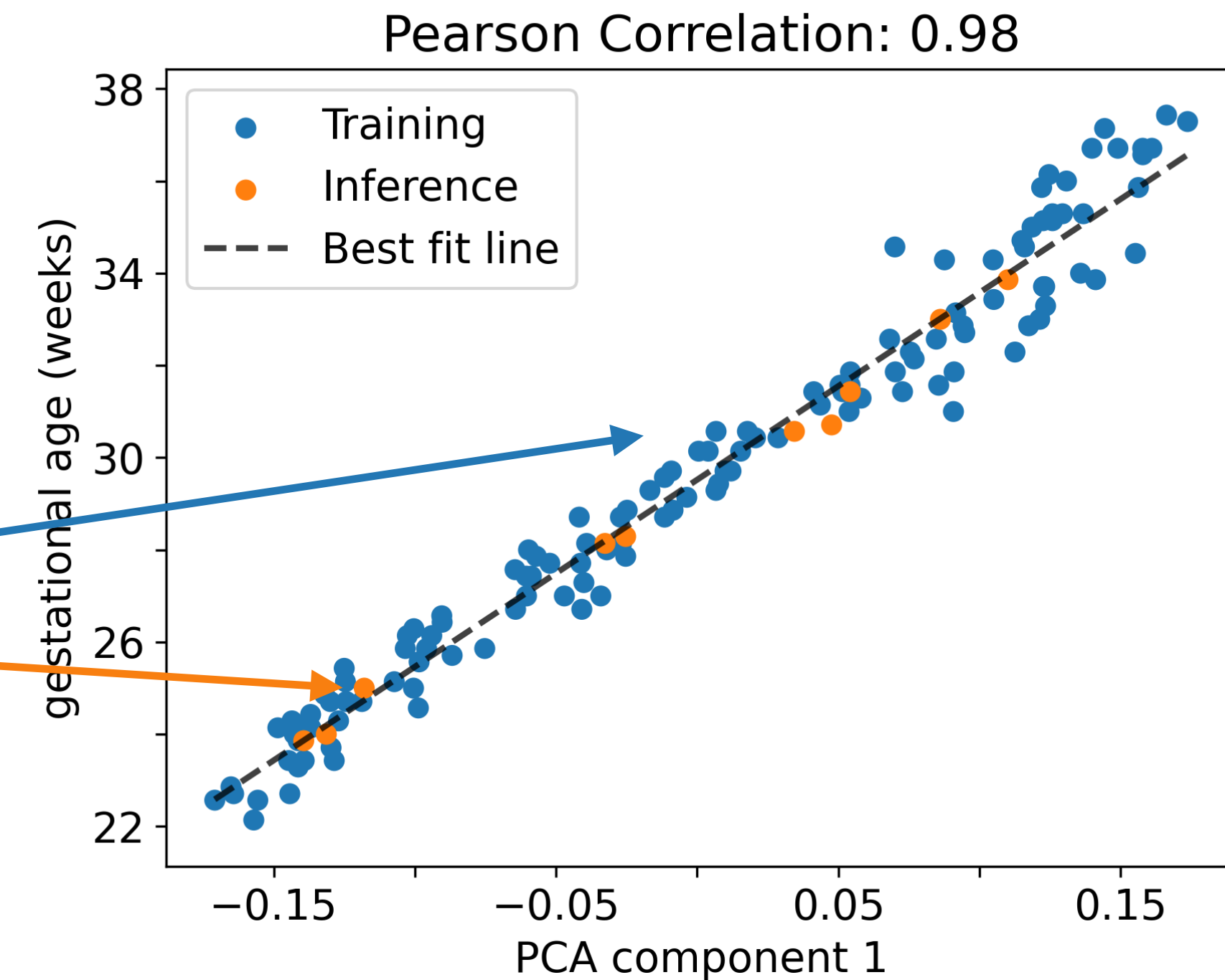
4) Ground Truth



Latent Representation



PCA



Age prediction (MAE): 0.23 ± 0.21 weeks

Conclusion

Classic Atlases:

- capture **discrete time-points** (weeks) with **discrete spatial-resolution**.
- constitute a **static representation** of the (healthy) fetal brain.
- often **fail to** adequately **model pathological brains**.

Conditional Implicit Neural Atlas (CINA):

- **represents** the fetal brain for the **entire time trajectory**.
- generates atlases **continuous in time and space**.
- models **neurotypical** and **pathological** brains by anatomical **conditioning**.

QUESTIONS

Latent Representation

